

Tan Kai Xun Dave

+65 9679 7951 | davetankx@gmail.com | [linkedin.com/in/tan-kai-xun-dave](https://www.linkedin.com/in/tan-kai-xun-dave) | github.com/Fasnon | Singapore Citizen

EDUCATION

- National University of Singapore** 2026 – Expected 2030
Bachelor of Computing in Computer Science (Incoming)
– NUS Merit Scholarship; NUS College (Honours Programme)
- NUS High School of Mathematics and Science** Jan 2018 – Dec 2023
NUS High School Diploma with High Distinction (CAP: 4.9/5.0)
– Majored in Chemistry (Honours), Mathematics (Honours), Computer Science, and Physics
– ExCo: Mathematics Interest Group (IG); Member: Chemistry, CS, and Humanitarian Education IGs
– Achieved **Gold** — Singapore Chemistry Olympiad and **Silver** — Singapore Mathematical Olympiad

EXPERIENCE

- Cyber Operator, Cyber Threat Intelligence Department** Apr 2024 – Present
The Digital and Intelligence Service (DIS) Singapore
– Monitored and assessed threats to Singapore's cyberspace; produced daily reports for dissemination.
– Identified and spearheaded AI optimisation opportunities in intel collection flows reducing manning needs.
– Revamped data pipelines from various intel sources to significantly improve IOC triage speed, accuracy and depth.
– Awarded **Best in Knowledge** and **Top Trainee** – 18/24 Cyber Defence Operator Course (CDOC).
- Project Intern, Data and Analytics Enablement, Digital Innovation** Oct 2023 – Mar 2024
Temasek Holdings Singapore
– Spearheaded development of process discovery framework and web frontend on production databases.
– Optimised automated workflows, cutting runtime/memory by 70%
– Streamlined account migration to service accounts, improving system reliability, security, and operational uptime.
– Prepared technical documentation and product showcases to key stakeholders.
- Research Attachment, C4NGP** Jun 2021 – Dec 2021
National University of Singapore Singapore
– Modelled constrained robust optimisation problem for berth allocation and quay crane scheduling using linear programming; proposed robust solutions to mentors.

PROJECTS

- Singapore Informatics League** | *Vue.js, Firebase, Tailwind CSS, GitHub* 2025 – Present
– Full stack web developer for the Singapore Informatics League.
– Supported inaugural event (Nov 2025): 620 players across 150 teams; handled 14,788 submissions in 4-hour contest.
– Developed responsive UI/UX, real-time scoreboard, registration portal, and backend integrations for announcements/key dates; ensured smooth high-traffic performance.
- Capture-The-Flag (CTF) Challenges** | *Cybersecurity, Cryptography, Forensics* 2022 – Present
– Achieved **1st** – BlahajCTF 2023, **2nd** – Lag n Crash 2024, **3rd** – Cyberthon 2023
– Solved AI/ML, reverse engineering, crypto, and web exploitation challenges with Python, SageMath, and Ghidra.
– Authored detailed challenge write-ups, earning recognition for GreyCTF 2025, Dice Quals 2024
- FiggleSpeak** | *PyTorch, Audio Processing, ffmpeg, Vue.js, Docker, HuggingFace* 2024
– AI app providing real-time pronunciation feedback from live audio.
– Integrated AI inference pipeline into backend APIs and docker-containerised cloud infrastructure.
- Tour Booking Website** | *Vue.js, Firebase* 2022 – 2023
– Full-stack web application for NUS High Open House, managing over 3,000 concurrent visitors.
– Implemented real-time booking and authentication using Firebase services.
– Deployed and maintained platform in coordination with organisers; conducted training for volunteers.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript
Domains: Data Analysis, Machine Learning, Web Development, Cybersecurity, Cryptography, Cyber Threat Intelligence
Tools: Power Automate, Firebase, SageMath, NumPy, Pandas, React, Vue.js, SQL Databases, OR-Tools, Ghidra, Google Cloud
Soft Skills: Technical Communication, Optimisation, Data Modelling, Visualisation